

# AYMANE AIT DADS

AI Research Intern | LLM Fine-Tuning · Computer Vision · Deep Learning Research

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## PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by applying LoRA SFT to a 30B-parameter Mamba-Transformer, demonstrating graduate-level research execution on large-scale model fine-tuning. Ranked 1st in the EURECOM class leaderboard on AI-generated image detection (0.791 AUC) using CLIP ViT-L/14 with a custom MLP head, and submitted results to the NTIRE 2026 @ CVPR CodaBench international benchmark. Brings end-to-end research competencies across deep learning, computer vision, NLP, and ablation-driven experimentation using PyTorch, Hugging Face Transformers, and PEFT on large-scale GPU workloads. Aims to contribute to CYENS research initiatives by advancing interactive media and smart systems through rigorous experimental methodology and state-of-the-art model development.

## CORE COMPETENCIES

Deep Learning Research | LLM Fine-Tuning | Computer Vision | Ablation Studies | Transformer Models | NLP & Reasoning | Experimental Methodology | PyTorch & GPU Computing

## WORK EXPERIENCE

**Orange Maroc** Jul 2024 - Aug 2024

**Data Science Intern** - Casablanca, Morocco

- Designed and trained **Random Forest and K-Means models** on 100,000+ fiber-optic network records to classify network quality across thousands of nodes using upload speed, latency, and jitter features.
- Built an end-to-end clustering pipeline that mapped customer performance profiles to **5 commercial service tiers**, with outputs adopted directly by the network optimization team for production decisions.
- Reduced manual network analysis time by **~60%** through automated feature engineering, model evaluation workflows, and reproducible Python pipelines built with Pandas and Scikit-learn.
- Delivered stakeholder presentations translating **model outputs into actionable maintenance recommendations**, bridging technical deep learning results with business decision-making using Power BI dashboards.

## PROJECTS

**NVIDIA Nemotron Model Reasoning Challenge** Kaggle · In Progress (June 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context).
- Performed **task-level error analysis** across 5 reasoning categories (unit conversion, cipher, symbolic, gravity, bit manipulation); built a diagnostic evaluation pipeline revealing that reasoning over-expansion - not incorrect reasoning - was the primary failure mode, and built custom cipher parsers and 8-bit operation solvers to produce verified chain-of-thought training data.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Kaggle GPU

**AI-Generated Image Detection** EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in the EURECOM class private Kaggle leaderboard** (0.791 AUC) by fine-tuning CLIP ViT-L/14 (428M parameters) with a custom MLP head and dual learning-rate strategy (2e-6 backbone, 5e-4 head) across 250K images from 25+ generator types; also submitted independently to the **NTIRE 2026 @ CVPR CodaBench** international benchmark.
- Discovered that **1-epoch training (0.793 AUC) outperformed 4-epoch training (0.767 AUC)** for out-of-distribution generalization; applied a 3-model ensemble weighted by validation AUC with 10-view test-time augmentation including flips, crops, blur, and resizes.

PyTorch · OpenCLIP · Hugging Face Transformers · FP16 Mixed Precision · Kaggle GPU

**Anomalous Sound Detection - Industrial Equipment** EURECOM · 2025 · Unsupervised Fault Detection

- Built an unsupervised fault detection system for slide-rail industrial machinery achieving **AUC 0.80** for predictive maintenance without any labeled anomaly examples available during training.
- Designed a **Transformer autoencoder** operating on Mel spectrograms with SpecAugment data augmentation, applying deep learning methodology to signal processing for smart systems applications.

PyTorch · Transformer Autoencoder · Mel Spectrograms · SpecAugment

## EDUCATION

**Engineering Degree in Data Science (Master-Level)** - EURECOM Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Image Security, Statistics, Cloud Computing, Web Applications

**CPGE - Mathematics & Physics** - Ibn Timiya 2021 - 2023

## SKILLS

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**LLM Fine-Tuning & Reasoning:** LoRA · QLoRA · SFTTrainer · PEFT · Unsloth · TRL · vLLM · chain-of-thought · Mamba-Transformer

**Deep Learning & Computer Vision:** PyTorch · Hugging Face Transformers · OpenCLIP · CLIP ViT · CNN · DenseNet · TensorFlow · Scikit-learn

**MLOps & Data Science:** Python · Git · FP16/BF16 mixed precision · Pandas · NumPy · Triton · CUDA · Power BI | SQL